SPLUNK LABS ENVIRONNEMENT

Write-Up

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[FIRST-TIME LAB SETUP]

Setting up a lab environment for future labs

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Goal

Objective

The main objective of this venture is to have four machines within the same network:

- A service server
- A SIEM Server/Log aggregation server
- A client
- An attacker

The virtual machines must be able to connect to each other and the internet.

Considerations

Four virtual machines might be a bit much for the average computer. That is why I have a server set up, mainly for this purpose. I must also ensure that the service ports I open do not get forwarded outside of the virtual LAN to avoid complications with outside attackers. I am stating this because there is a high chance I will purposefully make some services vulnerable for pen-testing and research purposes in the future.

Expectations

I expect things to go smoothly. I have significant experience with virtualisation and networking. The setup should be operational rather quickly.

Methodology

Instead of installing OSes from scratch with ISO images, I will be using tools to make the process quicker.

Tools

- Vagrant
- libvirtd, virsh & QEMU/KVM
- Virtual Machine Manager (Client APP)

Using the QEMU/KVM protocol, I will be able to manage and access virtual machines Vagrant will have created for me (as user session VMs). The management of a virtual network will be easy and convenient from there thanks to the GUI. For the creation of the network itself, however, I will use the CLI.

Once the network connections are established. It’s only a matter of installing Splunk on the SIEM Server, and forwarders on both the service server and client.

Operating Systems

Machine	OS	Why?
Service Server	Ubuntu	Industry Standard for web servers
SIEM Server	Debian	Stable and LTS
Client	Windows 10	Most clients use Windows
Attacker	Kali Linux	Pre-loaded with tools

Execution

Getting the VMs to post

```
arcade@dev-arcade:~/vms/splunk_env$ mkdir ubuntu_service
arcade@dev-arcade:~/vms/splunk_env$ mkdir siem_debian
arcade@dev-arcade:~/vms/splunk_env$ mkdir client_win10
arcade@dev-arcade:~/vms/splunk_env$ mkdir attack_kali
```

Figure 1: Creating of directories for vagrant files.

Since I'm using Vagrant, I can use pre-made boxes I can find on the [Vagrant Public Registry](). Since I use libvirtd, I am seeking boxes that support that provider in particular.

```
arcade@dev-arcade:~/vms/splunk_env$ cd ubuntu_service
arcade@dev-arcade:~/vms/splunk_env/ubuntu_service$ vagrant init  bento/ubuntu-24.04
A `Vagrantfile` has been placed in this directory. You are now
ready to `vagrant up` your first virtual environment! Please read
the comments in the Vagrantfile as well as documentation on
`vagrantup.com` for more information on using Vagrant.
arcade@dev-arcade:~/vms/splunk_env/ubuntu_service$ vagrant up
Bringing machine 'default' up with 'libvirt' provider...
==> default: Box 'bento/ubuntu-24.04' could not be found. Attempting to find and install...
    default: Box Provider: libvirt
    default: Box Version: >= 0
==> default: Loading metadata for box 'bento/ubuntu-24.04'
    default: URL: https://vagrantcloud.com/api/v2/vagrant/bento/ubuntu-24.04
==> default: Adding box 'bento/ubuntu-24.04' (v202508.03.0) for provider: libvirt
    default: Downloading: https://vagrantcloud.com/bento/boxes/ubuntu-24.04/versions/202508.03.0/providers/lib
virt/amd64/vagrant.box
Progress: 2% (Rate: 22.2M/s, Estimated time remaining: 0:02:14)
```

Figure 2: Vagrant downloading and setting up the image

Once the download is finished, the machine will boot on its own, and I will be able to access it through VMM (Virtual Machine Manager) from my work computer.

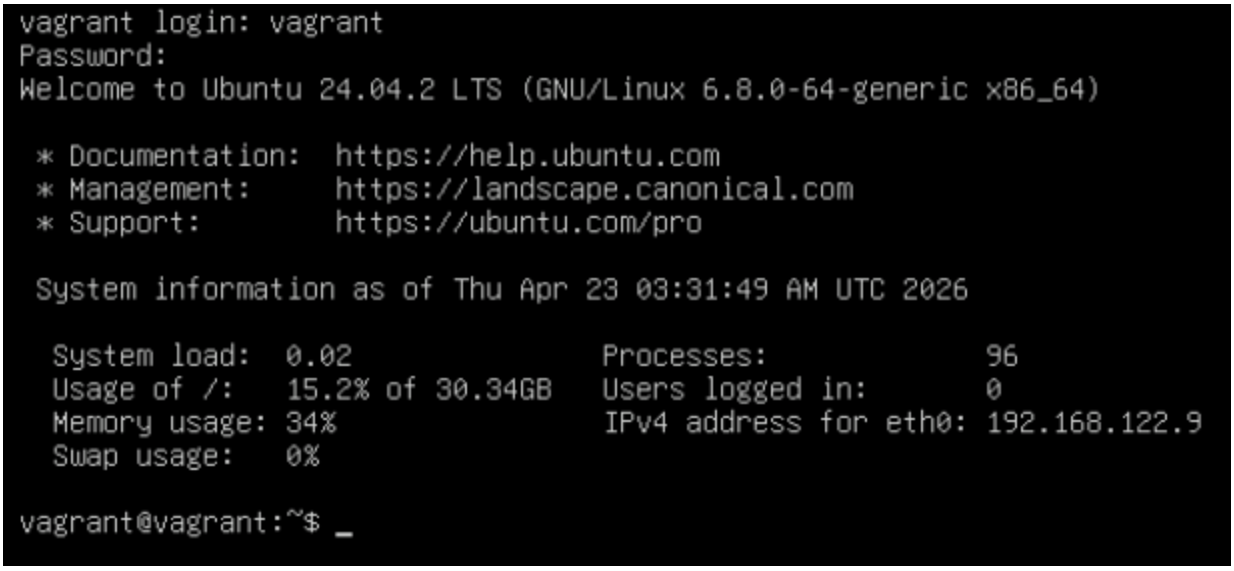


Figure 3: Ubuntu Web Server running

I just have to repeat these steps for the three other OSes now.

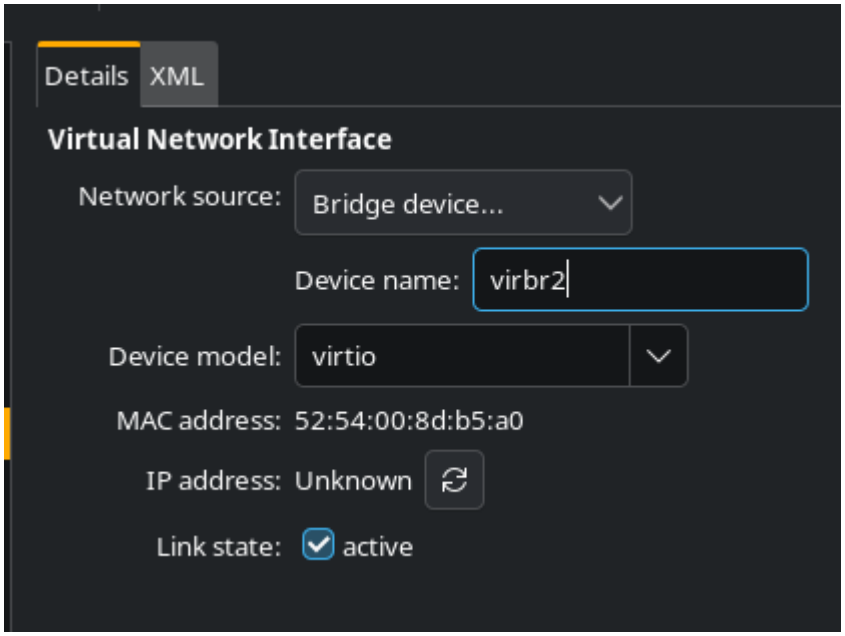
Networking

Now that all my machines are up and running, I must ensure they can locally reach others on their own LAN. Here's the XML for my custom LAN network.

```
<network>
  <name>splunk_lan</name>
  <forward mode="nat"/>
  <bridge name="virbr2" stp="on" delay="0"/>
  <ip address="192.168.100.1" netmask="255.255.255.0">
    <dhcp>
      <range start="192.168.100.100" end="192.168.100.200"/>
    </dhcp>
  </ip>
```

</network>

I need to assign the machines to the right network bridge now.



And finally, allow the bridge to be built by QEMU.

```
arcade@dev-arcade:~/vms/splunk_env$ nano network.xml
arcade@dev-arcade:~/vms/splunk_env$ cat /etc/qemu/bridge.conf
allow virbr0
allow eno1
arcade@dev-arcade:~/vms/splunk_env$ sudo nano /etc/qemu/bridge.conf
arcade@dev-arcade:~/vms/splunk_env$ cat /etc/qemu/bridge.conf
allow virbr0
allow virbr2
allow eno1
arcade@dev-arcade:~/vms/splunk_env$
```

And we’ve got our network.

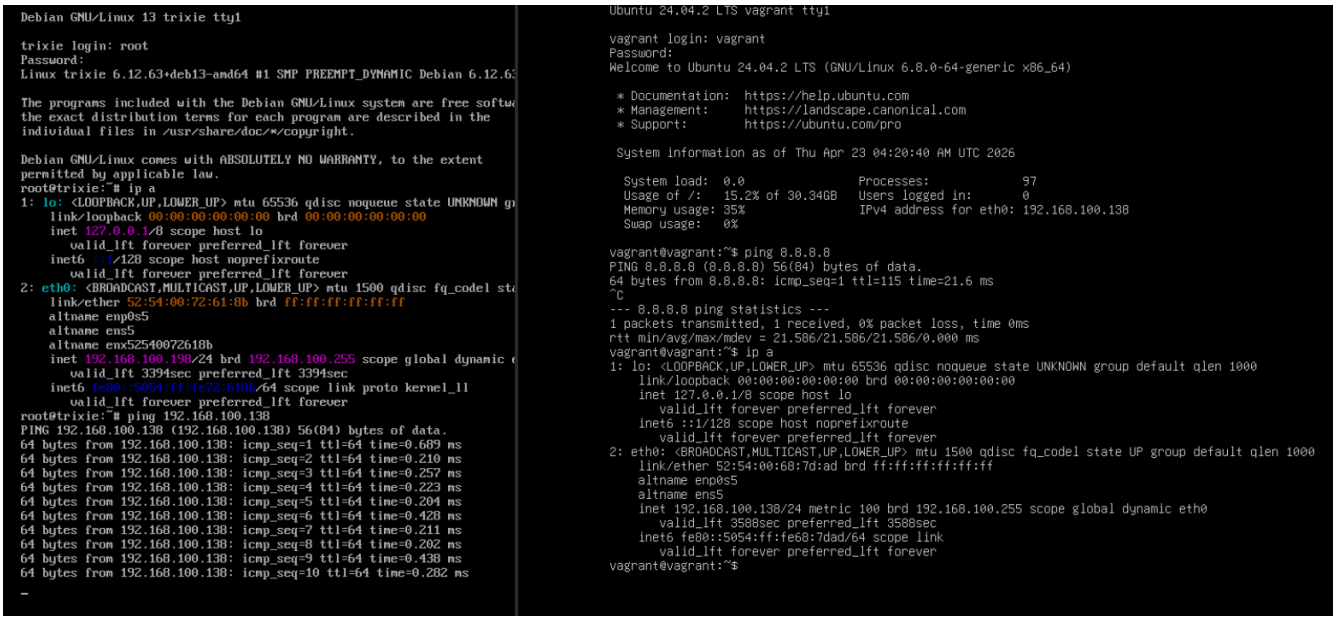


Figure 4: The SIEM Server is pinging the web server

Web Server Dummy Setup

For the Ubuntu machine, I will setup a base HTTP-only NGINX service. It is rather simple in concept. I changed the NGINX config to show a dummy page and started the service.

I ran into a permission issue because I decided to write the HTML code under a /home/ folder, but I quickly realized my mistake and moved it to a proper directory where I could manage permissions better. This ensures I won’t be exposing sensitive data in the future.

SIEM Server Splunk Initial Setup

I’ll be using the trial of Splunk Enterprise for now. I am aware of its data limitation of 500MB per day, but I will deal with it once

the problem arises. The password for the admin user will be *VagrantMachine*.

Just had to install a .deb, setup a Splunk user to run the binary...aaaand we are in!

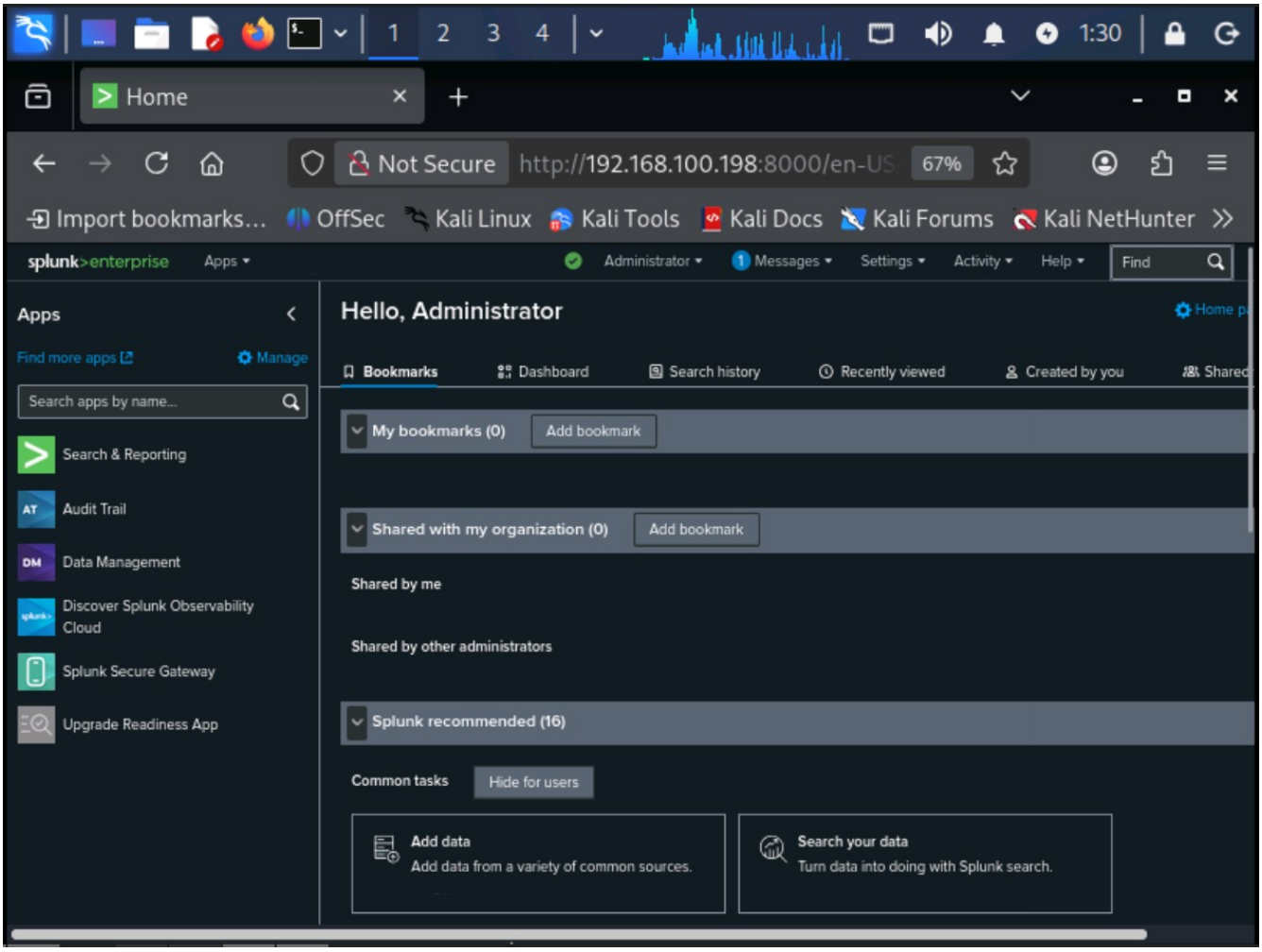
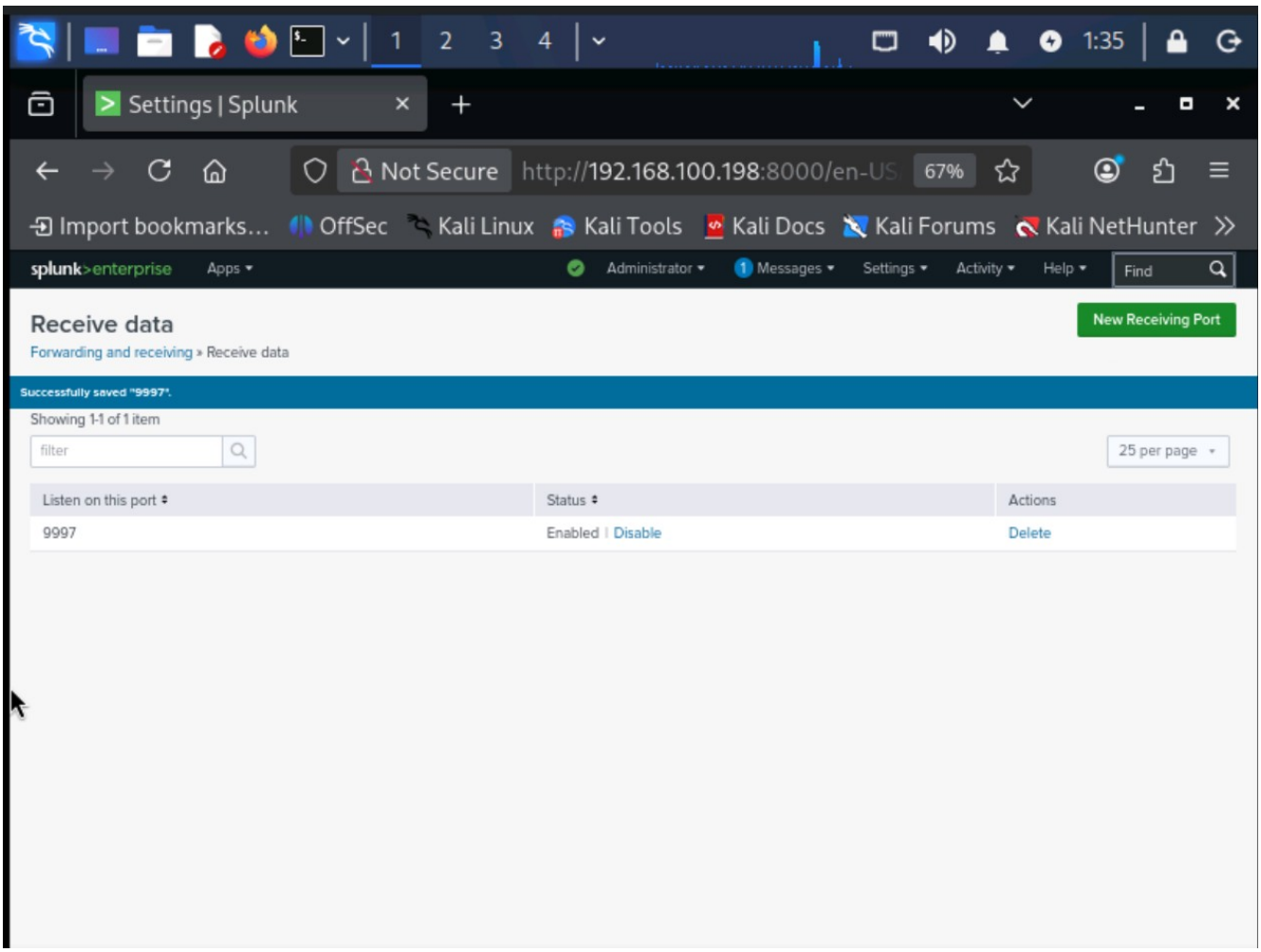


Figure 5: Splunk Web Interface

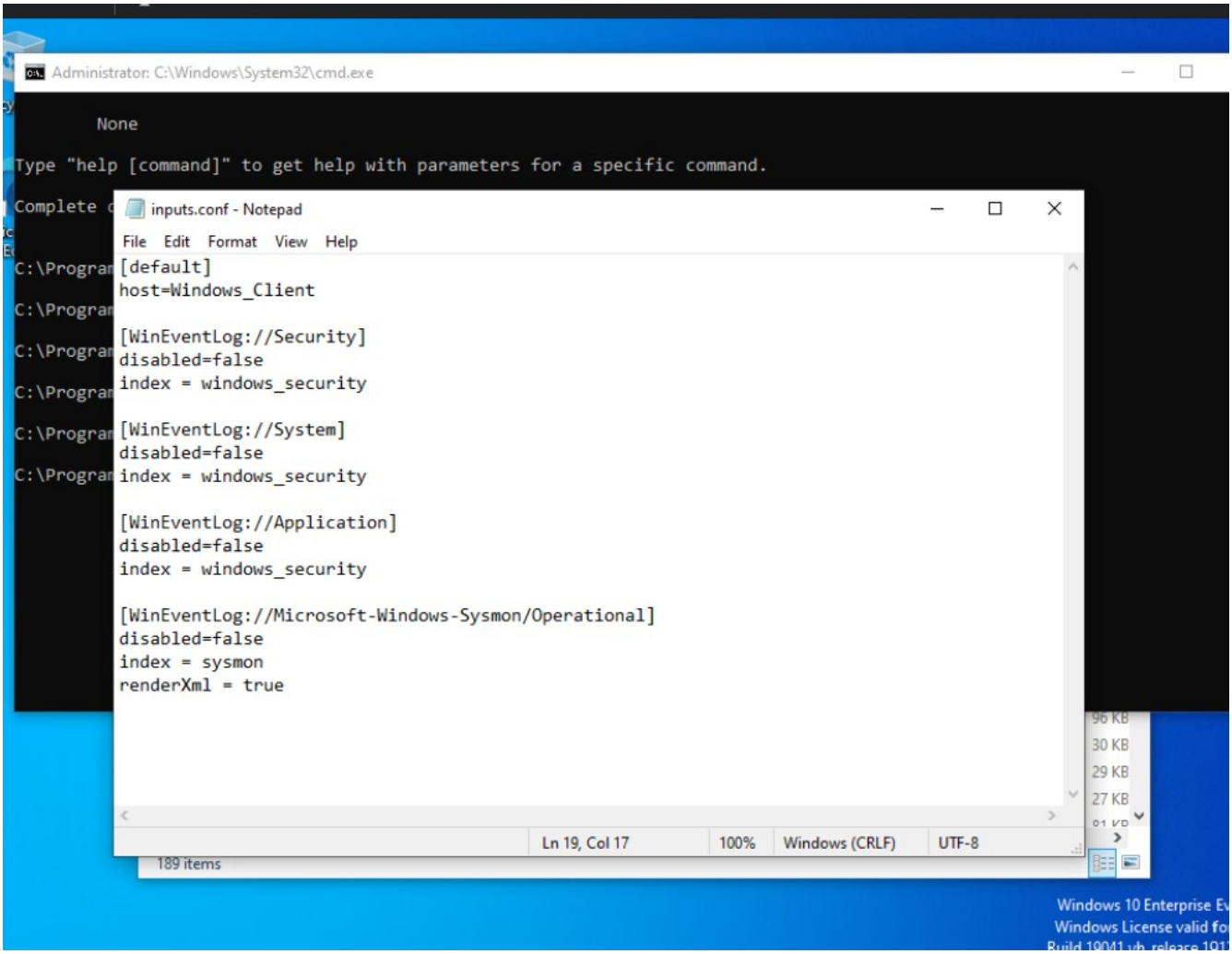
There was no firewall on the Debian machine (we'll have to fix that in another LAB), and Splunk listened from all available interfaces for the web interface, so I just accessed it from another machine on the virtual LAN: the attacking machine.

Forwarding Logs

Our SIEM server will be listening on port 9997 for logs.



We must install Splunk Forwarder on the web server and the Windows 10 client, and then tell them what to send with inputs.conf.

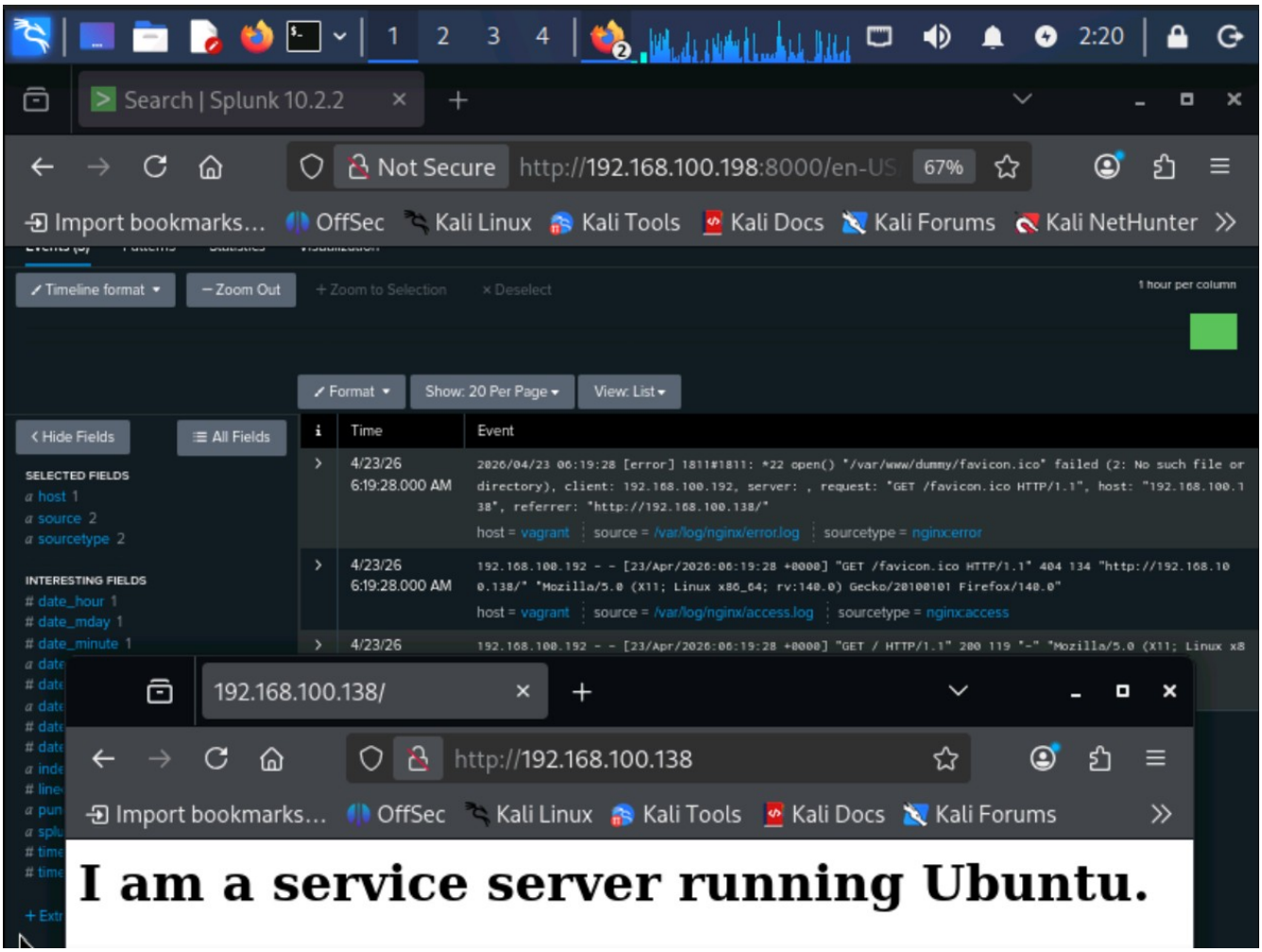


I will be sending the typical Windows system logs and the NGINX love to Splunk. I will add more logs to forward from the web server once I set up more services.

```
GNU nano 7.2                    /opt/splunkforwarder/etc/system/local/inputs.conf
[monitor:///var/log/nginx/access.log]
disabled = false
sourcetype = nginx:access
index = web

[monitor:///var/log/nginx/error.log]
disabled = false
sourcetype = nginx:error
index = web
```

I had actually forgotten to create the indexes I was pointing to in the inputs.conf, so I was not getting any logs. I troubleshooted for a dozen of minutes thinking it was something wrong with my network, but nope. Always create the indexes in advance, or else the logs will not be registered.



So there we have it now. Two machines sending their logs to our SIEM server!

Conclusion

I have done something similar in the past in college, but this is my first time setting it up from scratch myself on my home lab. I'm pretty happy with what I've learned while writing this report: more virsh, KVM XML, and to configure indexes BEFORE trying to fill them.

My next idea is to pen-test my own web server, and see how traceable I will be through the logs. Afterwards, I can recognize attack

patterns and create alerts based on that... and eventually, set up a SOAR system. So much LAB potential with these babies... it makes my head spin! This is only the very start of my adventure with this setup, and I hope you will read my other reports to see where I go from here.